# AQ10.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.
**Level 1:**

Consider a function $f: R^2 \to R$f : R2 → R defined as:

$$f(x, y)\text{f(x, y)} = \begin{cases} x^2 \sin\frac{1}{x} + y & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}\{\text{x2 sin x1 + y0if x} \quad = 0\text{if x = 0.}$$

- **Statement 1:** $f_y$ fy is continuous at the origin.
- **Statement 2:** $f_x$ fx is continuous at the origin.

Enter the number of correct statements.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 0
*1 point*
*1 point*
A function $f: R^2 \to R$f : R2 → R satisfies that $f_x(x, y) = c$fx(x, y) = c and $f_y(x, y) = k$fy (x, y) = k of all points of $R^2$ R2, where $c$c and $k$k are some constants, then which of the following option(s) is (are) true?

☐ The tangent line exists at any point of $R^2$ R2 in any direction.
☐ The tangent lines at the origin do not lie on a plane.
☐ There exist points in $R^2$ R2 for which tangent points in all directions do not exist.
☐ $f(x, y)$f(x, y) can be $f(x, y) = cx + ky + \alpha$f(x, y) = cx + ky + α, where $\alpha$α is any real number.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
The tangent line exists at any point of $R^2$ R2 in any direction.
$f(x, y)$f(x, y) can be $f(x, y) = cx + ky + \alpha$f(x, y) = cx + ky + α, where $\alpha$α is any real number.
*1 point*
Consider $f: R^2 \to R$f : R2 → R and suppose partial derivatives of $f(x, y)$f(x, y) with respect to $x$x and $y$y are continuous at a point $(a, b)$(a, b). Then which of the following option(s) is (are) true?

☐ $f_x(a, b) = \lim_{(x, y) \to (a, b)} f_x(x, y)$fx(a, b) = $\lim_{(x,y)\to(a,b)}$fx(x, y)
☐ $f_y(a, b) = \lim_{(x, y) \to (a, b)} f_y(x, y)$fy(a, b) = $\lim_{(x,y)\to(a,b)}$fy(x, y)
☐ $f_x(a, b) \neq \lim_{(x, y) \to (a, b)} f_x(x, y)$fx(a, b) $\quad = \lim_{(x,y)\to(a,b)}$fx(x, y).
☐ $f_y(a, b) \neq \lim_{(x, y) \to (a, b)} f_y(x, y)$fy(a, b) $\quad = \lim_{(x,y)\to(a,b)}$fy(x, y)

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$f_x(a,b) = \lim_{(x,y)\to(a,b)} f_x(x,y)$fx(a, b) = (x,y)→(a,b)lim fx(x, y)

$f_y(a,b) = \lim_{(x,y)\to(a,b)} f_y(x,y)$fy(a, b) = (x,y)→(a,b)lim fy(x, y)

*1 point*
Consider a function $f(x,y) =$ f(x, y) = $x^2 + y^2$ x2 + y2. Which of following is the tangent at the point (1,2) in the direction of (1, 1)?

○ $x = 1 + \frac{t}{\sqrt{2}}, y(t) = 2 + \frac{t}{\sqrt{2}}, z(t) = 5 + \frac{6t}{\sqrt{2}}$x = 1 + 2 √ t, y(t) = 2 + 2 √ t, z(t) = 5 + 2 √ 6t

○ $x = 1 + t, y(t) = 2 + t, z(t) = 5 + 6t$x = 1 + t, y(t) = 2 + t, z(t) = 5 + 6t

○ $x = 1, y(t) = 2 + t, z(t) = 5 + \frac{6t}{\sqrt{2}}$x = 1, y(t) = 2 + t, z(t) = 5 + 2 √ 6t

○ $x = 1 + t, y(t) = 2 + t, z(t) = 4 + \frac{6t}{\sqrt{2}}$x = 1 + t, y(t) = 2 + t, z(t) = 4 + 2 √ 6t

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$x = 1 + \frac{t}{\sqrt{2}}, y(t) = 2 + \frac{t}{\sqrt{2}}, z(t) = 5 + \frac{6t}{\sqrt{2}}$x = 1 + 2 √ t, y(t) = 2 + 2 √ t, z(t) = 5 + 2 √ 6t

**Level 2:**

*1 point*
Consider a function

$$f(x,y) = \begin{cases} \frac{x}{y} \text{if} y \neq 0 \\ 0 \text{if} y = 0. \end{cases}$$.f(x, y) = {yx0if y = 0if y = 0. .

Which of the following is the tangent line passing through the point (1, -1) and parallel to the line

$x(t) = \frac{t}{\sqrt{5}}, y(t) = 1 + \frac{2t}{\sqrt{5}}, z(t) = -\frac{3t}{\sqrt{5}}$x(t) = 5

√

t, y(t) = 1 + 5

√

2t, z(t) = − 5

√

3t?

○ $x(t) = 1 + t, y(t) = -1 + 2t, z(t) = -\frac{3}{\sqrt{5}}t$x(t) = 1 + t, y(t) = −1 + 2t, z(t) = − 5

√

3t

○ $x(t) = 1 + t, y(t) = 1 + 2t, z(t) = -1 - \frac{3}{\sqrt{5}}t$x(t) = 1 + t, y(t) = 1 + 2t, z(t) = −1 − 5

√

3t

○ $x(t) = 1 + \frac{t}{\sqrt{5}}, y(t) = -1 + \frac{2t}{\sqrt{5}}, z(t) = -1 - \frac{3t}{\sqrt{5}}$x(t) = 1 + 5

√

t, y(t) = −1 + 5

√

2t, z(t) = −1 − 5

√

3t

○ $x(t) = -1 + t, y(t) = -1 + 2t, z(t) = -1 - \frac{3}{\sqrt{5}}t$x(t) = −1 + t, y(t) = −1 + 2t, z(t) = −1 − 5

√

3t

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$x(t) = 1 + \frac{t}{\sqrt{5}}, y(t) = -1 + \frac{2t}{\sqrt{5}}, z(t) = -1 - \frac{3t}{\sqrt{5}}$x(t) = 1 + 5

√

t, y(t) = −1 + 5

√

2t, z(t) = −1 − 5

√

3t

*1 point*

Consider a function $f(x, y, z) = 2x^2 + xy + z^2$ f(x, y, z) = 2x2 + xy + z2. Which of following is the tangent at the point (1,1,1) in the direction of (1, 0, 0)

○ $(1, 1, 1, 4 + 5t)$(1, 1, 1, 4 + 5t)

○ $(1 + t, 1, 1, 4 + t)$(1 + t, 1, 1, 4 + t)

○ $(1, 1 + t, 1 + t, 4 + 5t)$(1, 1 + t, 1 + t, 4 + 5t)

○

$(1 + t, 1, 1, 4 + 5t)$(1 + t, 1, 1, 4 + 5t)

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$(1 + t, 1, 1, 4 + 5t)$(1 + t, 1, 1, 4 + 5t)

*1 point*

Consider the following functions $f$ f defined as:

$$f(x, y, z) = xyz.\text{f(x, y, z) = xyz.}$$

Then the tangent line at the point (1,0,1) in the direction of (1,1,1) is parallel to the line

○ $x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$x(t) = 1 + 3

√

t, y(t) = 3

√

t, z(t) = 1 + 3

√

t, u(t) = 3

√

t

○ $x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = t\sqrt{3}$x(t) = 1 + 3

√

t, y(t) = 3

√

t, z(t) = 1 + 3

√

t, u(t) = t 3

√

○ $x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = 1 + \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$x(t) = 1 + 3

√

t, y(t) = 1 + 3

√

t, z(t) = 1 + 3

√

t, u(t) = 3

√

t

○ None of the above.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$x(t) = 1 + 3

$\sqrt{}$
t, y(t) = 3

$\sqrt{}$
t, z(t) = 1 + 3

$\sqrt{}$
t, u(t) = 3

$\sqrt{}$
t

Check Answers
Your score is: 0/7